

Introduction of Geographical Information System

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Objective of the course

- 1/ Basic knowledge on GIS: definitions, structural and functional components..
- 2/ Help students understanding the spatial information, spatial models and their organization
- 3/ Provide practical skills of using GIS software for some simple applications
- 4/ Think spatially: when and how to use certain operation/tools

Contents of the course

12 hours of lectures, 12 hours of practices

Attendance: 10%

Midterm; 40%

Final exam; 50%

(Group presentation)

(Paper test)

Chapter 1: What is Geographic Information System (GIS)?

Chapter 2: Spatial analysis

Chapter 3: Raster analysis

Chapter 4: Geospatial technologies

Textbooks

1/ *Paul Bolstad. 2016. GIS Fundamentals: A First Text on Geographic Information Systems. Eider Press, White Bear Lake, Minnesota*

2/ *Gretchen N. Peterson. 2009. GIS Cartography, A Guide to Effective Map Design. CRC Press, Taylor & Francis*

3/ *Michael Kenned. 2002. The Global Positioning System and GIS: An Introduction Second Edition, Taylor & Francis*

Chapter 1. What is Geographic Information System (GIS)?

1. Definition of GIS
2. Structure of GIS
3. Spatial models
4. Regression models

Reasons for GIS

- Our world is constantly changing, and not all changes are for the better.
 - *Natural causes: e.g., volcanic eruptions*
 - *Human causes: e.g., land use changes*
 - *Mix / Unclear causes: e.g., El Niño / La Niña events*
- We, humans, want to understand what is going on in our world, and to take actions.
- Requirement of understanding phenomena that have (a) a geographic dimension, and (b) a temporal dimension.
 - *Spatio-temporal: be of/in space and time*
- Space science and Satellite technology: answer the questions
 - *Structure of spatial data, temporal data, the way to integrate data, data in the coordinates*
 - *The final purposes of space science (Earth observation, simulations, distributions)*

“Everything that happens, happens somewhere in space and time.” – Michael Wegener
(University of Dortmund)

1/ Definition of GIS

GIS: a computer system is built to capture, store, manipulate, analyze, manage and display all kinds of spatial or geographical data.

GIS applications: allow end users to perform spatial query, analysis, edit spatial data and create maps.

GIS answers questions:

What appears in a location? (Mapping)

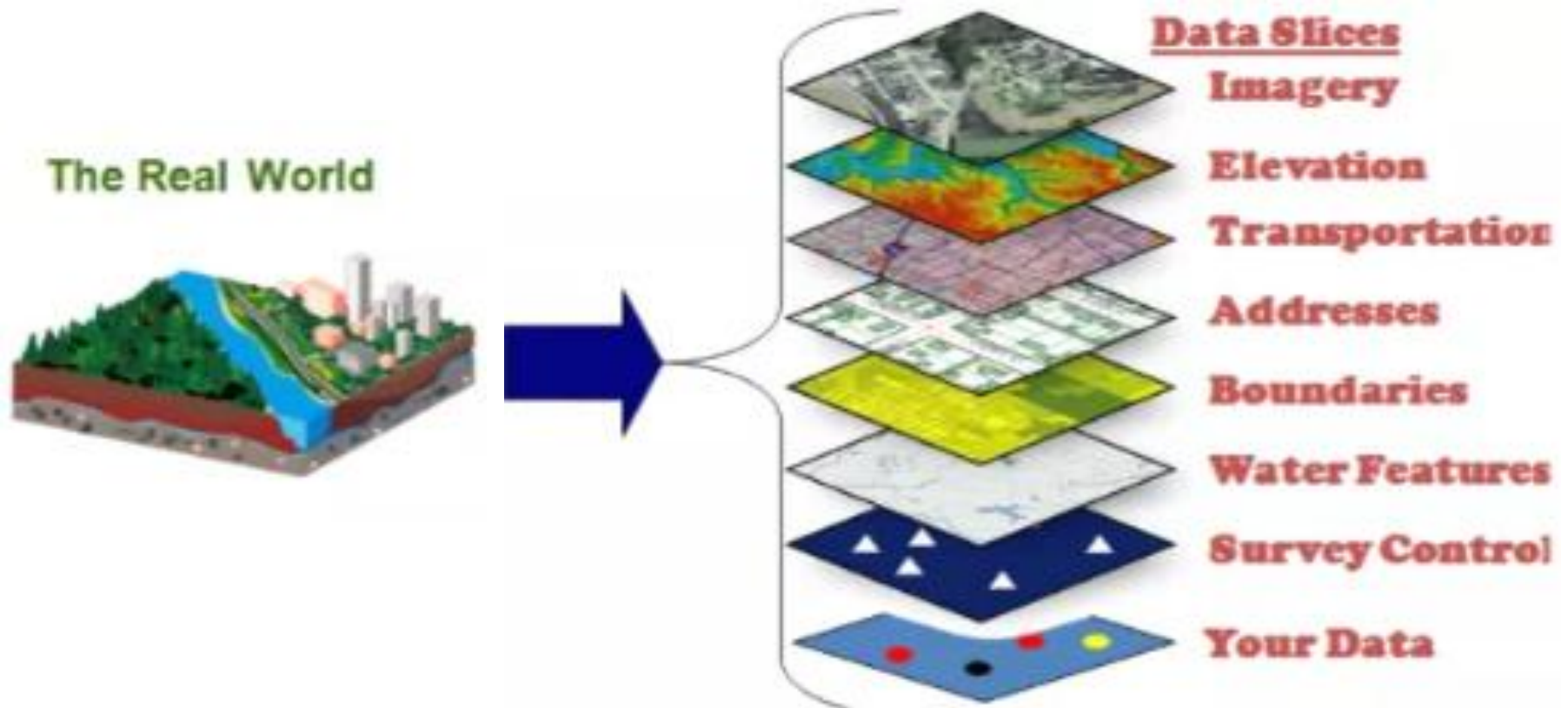
Where is a physical object located? (Spatial analysis)

When the phenomena occur? (Prediction GIS models)

Definition of GIS

- *A geographic information system (GIS) lets us visualize, question, analyze, and interpret data to understand relationships, patterns, and trends. (ESRI)*
- *In the strictest sense, a GIS is a computer system capable of assembling, storing, manipulating, and displaying geographically referenced information (that is data identified according to their locations). (USGS)*

Definition of GIS



GIS models to simulate the real world

$$\text{GIS} = \text{"G"} + \text{"I"} + \text{"S"}$$

- “G” = Geographic
 - Denotes the concept of spatial location on Earth’s surface
 - Importance of relative location (not just where you are but where you are in relation to everything else)
 - Theories and techniques in Geography form the basis of GIS

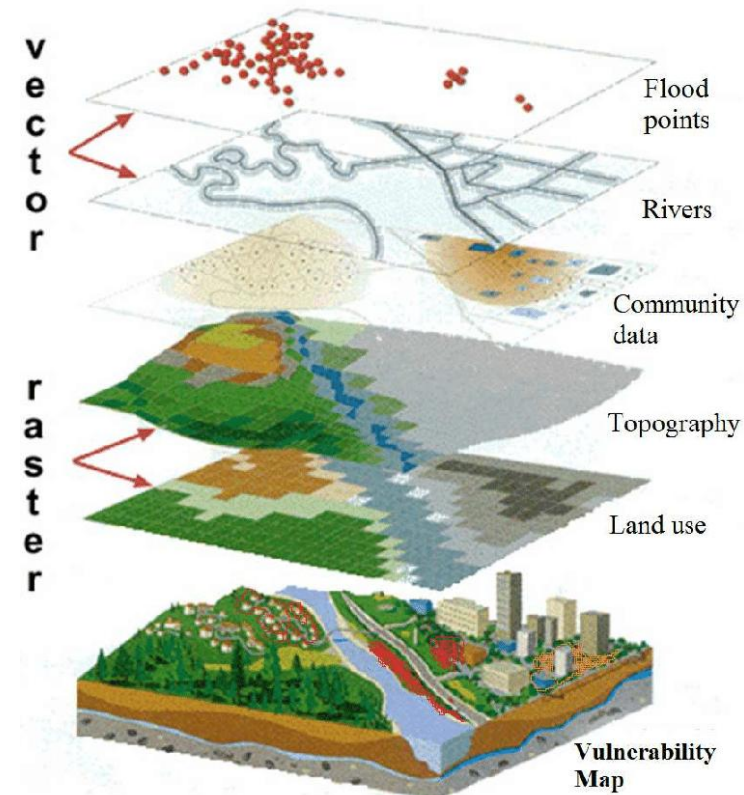
$$\text{GIS} = \text{“G”} + \text{“I”} + \text{“S”}$$

- “I” = Information
 - Substance (knowledge) about location
 - Factual and interpretative
 - Tables + Maps + Analysis
 - Transformation of table information into spatial context for analysis
 - Technology and computer systems
- “S”
 - Systems
 - Science
 - Studies
 - Services

Advantage of GIS

(No ways except GIS)

- Better, improving decision made by government people (the help of layered information)
- Citizen engagement due to better system
- Help to identify communities that is under risk or lacking infrastructure
- Helps in identifying criminology matters
- Better management of natural resources
- Better communication during emergency situation
- Cost savings due to better decision
- Finding different kinds of trends within the community
- Planning the demographic changes



Examples of GIS

- **Urban Planning, Management**
 - Land information acquisition
 - Economic development
 - Housing renovation programs
 - Emergency response
 - Crime analysis
- **Environmental Sciences**
 - Monitoring environmental risk
 - Modeling storm water runoff
 - Management of watersheds, floodplains, wetlands, forests
 - Environmental Impact Analysis
 - Hazardous or toxic facility siting
- **Political Science**
 - Analysis of election results
 - Predictive modeling
- **Civil Engineering/Utility**
 - Locating underground facilities
 - Coordination of infrastructure maintenance
- **Business**
 - Demographic Analysis
 - Market Penetration/ Share Analysis
- **Education Administration**
 - Enrollment Projections
 - School Bus Routing
- **Real Estate**
 - Neighborhood land prices
 - Traffic Impact Analysis
- **Health Care**
 - Epidemiology
 - Service Inventory

Geospatial technologies

- Geospatial technology / Geomatics
 - Land surveying
 - Remote sensing
 - Cartography
 - Spatial analysis
- – Global navigation satellite systems (GPS, GLONASS, Galileo, Compass)
 - Photogrammetry
 - Geography
 - ...

History of GIS

- **Year 1854:** Term of GIS, used points on London residential map to plot outbreak of Cholera



John Snow, Cholera Outbreak Map

- **Year 1960:** Modern computerized GIS system began in year 1960
- **Year 1962:** Dr. Roger Tomlinson (father of GIS): Canadian Geographic Information System (CGIS) to store, analyze and manipulate data. CGIS had the capacity to overlay, measurement and digitizing.
- **Year 1980:** GIS software: M&S Computing, Environmental Systems Research Institute (ESRI), Computer Aided Resource Information System (CARIS). ESRI products like ArcGIS, ArcView hold 80 % of global market.

Why do we need the GIS course in
the SST program?

2. Structure of GIS

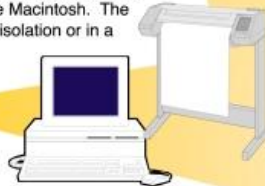
(GIS Components)

A Geographic Information System (GIS) links locational (spatial) and database (tabular) information and enables a person to visualize patterns, relationships, and trends. This process gives an entirely new perspective to data analysis that cannot be seen in a table or list format. The five components of a GIS are listed below.

HARDWARE

The hardware is the computer and peripherals on which the GIS operates. Today, this could be a centralized computer server running the UNIX or Windows NT operating systems, a desktop PC, or an Apple Macintosh. The computer may operate in isolation or in a networked configuration.

- Computers
- Networks
- Peripheral Devices
 - Printers
 - Plotters
 - Digitizers



SOFTWARE

GIS software provides the functions and tools users need to store, analyze, and display geographical information. The key software components are

- GIS Software
- Database Software
- OS Software
- Network Software



DATA

One of the most important component of GIS is the data. It is absolutely essential that data be accurate. The following are different data types:

- Vector Data
- Raster Data
- Image Data
- Attribute Data



GIS

PEOPLE

GIS technology is clearly of limited value without people to manage the system and to develop plans for applying it. Users of GIS range from highly qualified technical specialists to planners, foresters, and market analysts who use GIS to help with their everyday work.

- Administrators
- Managers
- GIS Technicians
- Application Experts
- End Users
- Consumers



METHODS

Methods are well designed plans and application-specific business rules describing how technology is applied. This includes the following:

- Guidelines
- Specifications
- Standards
- Procedures



Spatial data and Geoinformation in GIS

Data, Metadata, Spatial data, Geospatial data, information, Geoinformation

- **“Data”** are representations that can be operated upon by a computer.
- **“Metadata”** are data about data.
- **“Spatial data”** are data that contain positional values.
- **“Geospatial data”** are spatial data that are georeferenced.
 - In the context of GIS, spatial data and geospatial data are regarded as synonyms of georeferenced data.
- **“Information”** is the meaning of data as interpreted by human beings.
- **“Geoinformation”** is information that involves interpretation of spatial data.

The real world and representations of it

- Data and information are usually tried to represent some part of the real world as it is, as it was, or perhaps as we think it will be.
 - *We say ‘some part’ because the real world cannot be represented completely.*
- A computer, maps, papers representation of some part of the real world to enter and store data, analyze the data and transfer results to humans or to other systems.

A representation of some part of the real world can be considered a model of that part.

This allows us to study the model instead of the real world.

3. MODELS IN GIS

The real world and its spatial models

- **MODEL:** “A model is a manageable, comprehensible and schematic representation of a piece of reality”
 - “**reality**” – no hypothetical system
 - “**a piece of reality**” – limited domain in time and space
 - “**schematic representation**” – from a specific point of view
 - “**representation**” - an infinite number of projections
 - “**a comprehensible representation**” - a model serves a specific goal
 - “**a manageable representation**” - it should give the user the results they need

MODELS IN GIS

- Different types of models:
 - Most familiar is the **map**
 - A collection of stored data representing real-world phenomena is also a model—

data model

- Analogue maps
- Digital models



Paper map



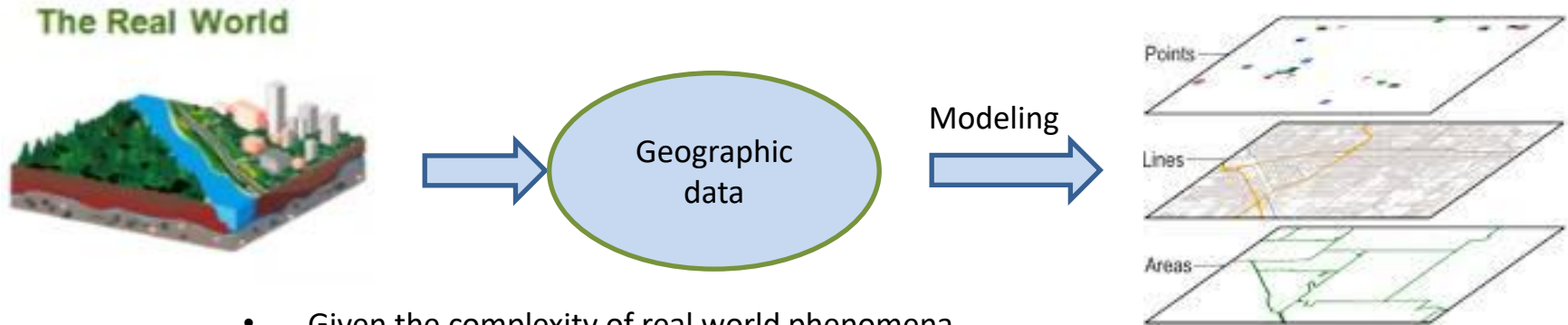
Computer representations



Cairo, Egypt

Characteristics of GIS models

What is the “problem” to be modeled? And How ?



- Given the complexity of real world phenomena
- Models can never be perfect.
 - Limitations on the amount of data that we can store
 - Limits on the amount of detail we can capture
 - Limits on the time we have available for a project.
- Some facts or relationships that exist in the real world may not be discovered through ‘models’
- Models come in many different flavors.
 - Maps, Databases..
- Most maps and databases can be considered static models.
- Dynamic models or process models address changes that have taken place, are taking place and may take place.

Characteristics of GIS models

(Scale in a digital model?)

- Spatial resolution/extent
- Temporal resolution/extent
- Define what is left out of the model
- Leave out uncertainty about model data, predictions
- Model must run faster than the real world

Characteristics of GIS models

What is the “topic” to be modeled?

What are the important phenomena?



Characteristics of GIS models

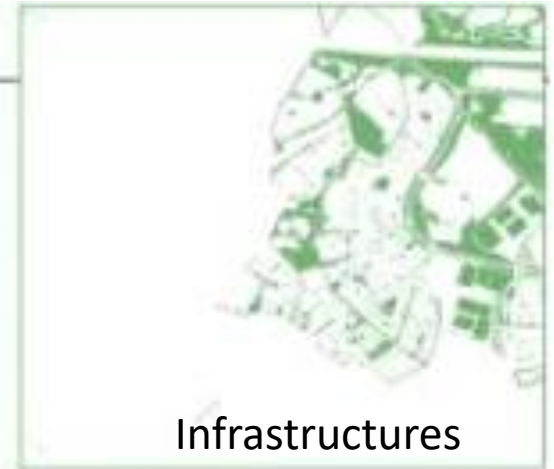
We model theme by determining the important phenomena.



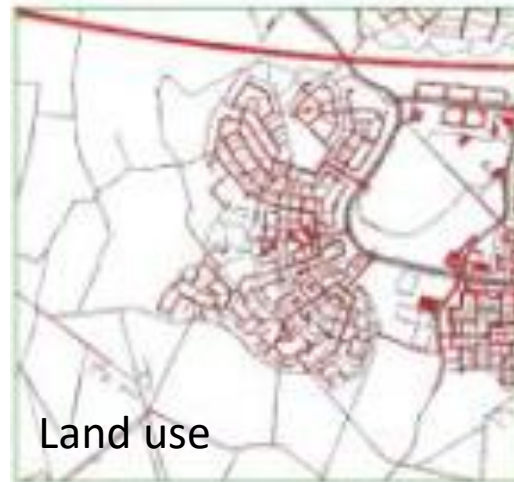
Topographical map scale
1:25.000



Buildings



Infrastructures



Land use



Water body

Characteristics of GIS models

What is the “scale” of the model?



Topographic Map 1:10 000



Topographic Map 1:50 000 after generalization

*Which are small scale, large scale?
What are the differences?*

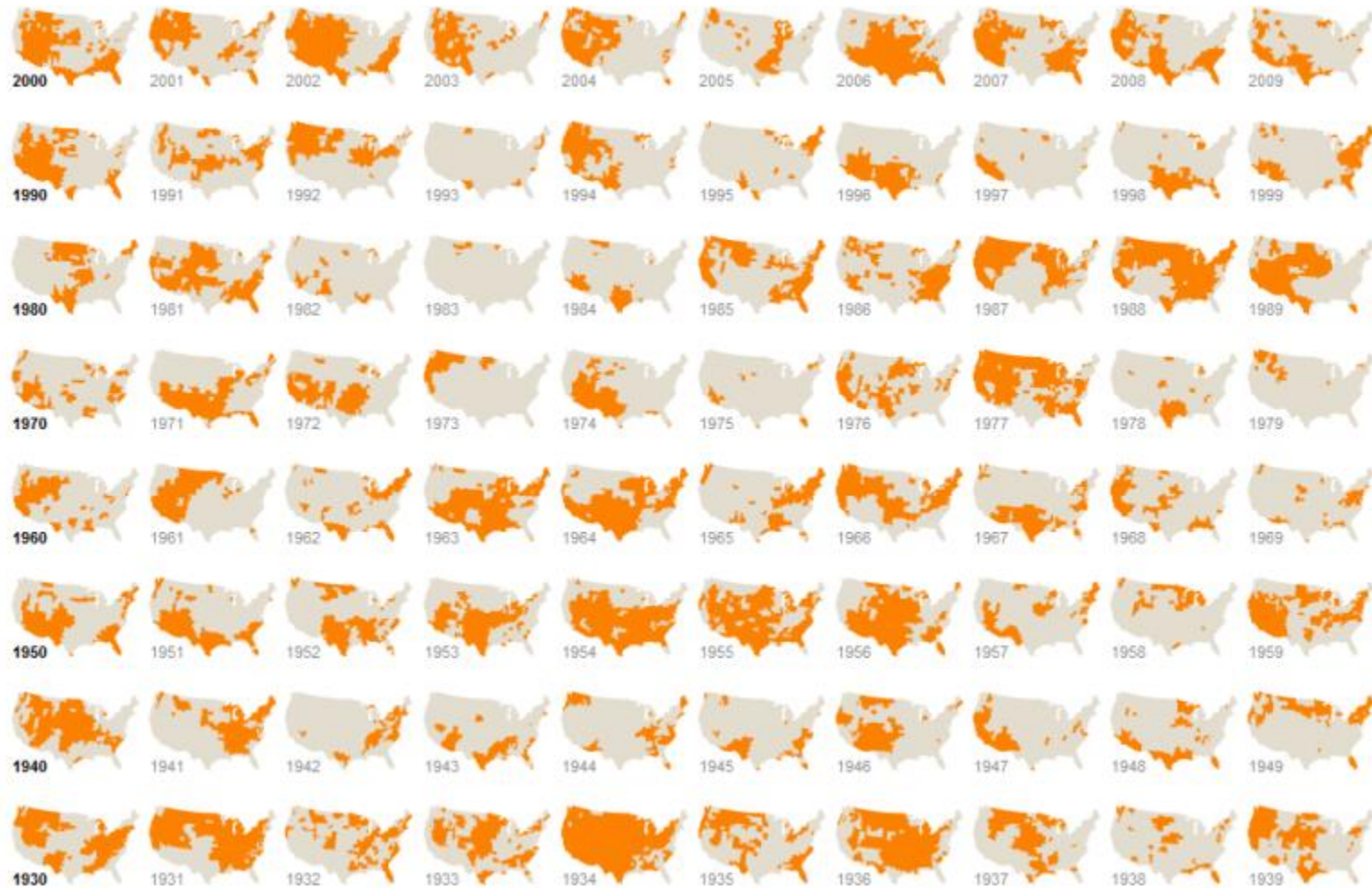
Characteristics of GIS models

Something are removed in the models?



Characteristics of GIS models

(Time scale of model)



Drought's Footprint (1930 to 2000)

Image source: National Climatic Data Center, NOAA

Paper maps Vs Digital model

- Fixed scale of printed map
- All layers in one
- Fixed coordinate system
- Uneditable
- Measurement limitation
- Area of interest is out of frame
- Unchangeable symbolization
- Mass production/printing
- Flexible scale selection (generalization)
- Storage of separate layers
- Projection
- Editable
- Spatial analysis function
- Selection of area of interest
- Symbolization
- Thematic map for single use

WHAT ARE GEOGRAPHIC PHENOMENA?

- A geographic phenomenon: a manifestation of an entity or process of interest:
 - Can be named or described
 - Can be geo-referenced
 - Can be assigned a time interval at which it is/was present
- Not all relevant information about phenomena has the form of a triplet:
 - No name (un-described object)
 - No geo-reference (legal document)
 - No time (phenomenon that exists permanently)

WHAT ARE GEOGRAPHIC PHENOMENA?

- **Euclidean space**

A GIS operates under the assumption that the relevant spatial phenomena occur in a two- or three-dimensional Euclidean space, unless otherwise specified.

Euclidean space can be informally defined as a model of space in which locations are represented by coordinates— (x, y) in 2D; (x, y, z) in 3D—and distance and direction can be defined with geometric formulas. In the 2D case, this is known as the Euclidean plane, which is the most common Euclidean space in GIS use.

TYPES OF GEOGRAPHIC PHENOMENA

Objects



Example:

- Building

Studied in groups:

- Building blocks

Characteristics:

- crisp boundaries
- Inside the boundary only one value

TYPES OF GEOGRAPHIC PHENOMENA

Fields: CONTINUOUS vs DISCRETE

- A field is a geographic phenomenon that *has a value 'everywhere'* in the study area.

- Mathematical function: $f(x, y)$

Stands for the value of the field “f” location (x,y)

- In continuous fields, the underlying function is assumed to be *'mathematically smooth'* - gradual change

- Continuous fields can be differentiable, meaning we can determine a measure of change in the *field value per unit of distance*

TYPES OF GEOGRAPHIC PHENOMENA

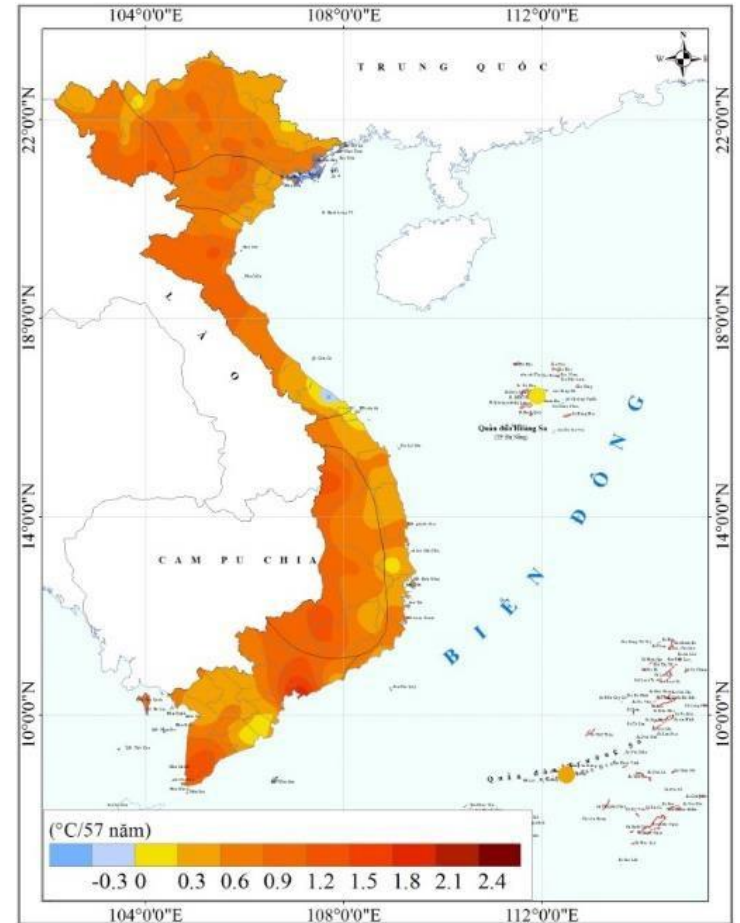
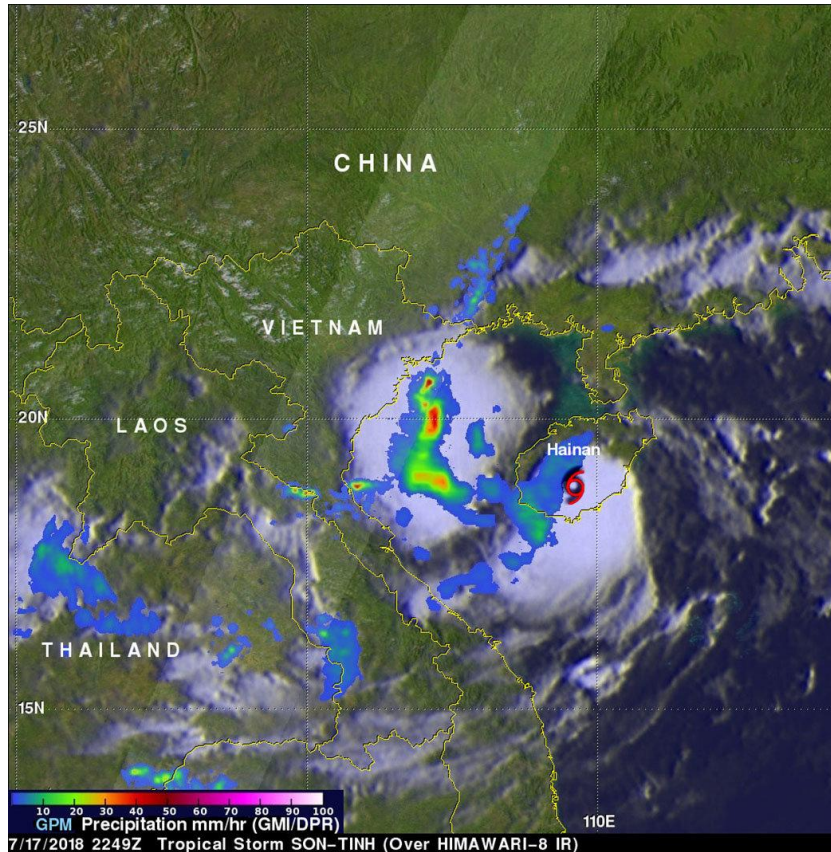
Fields: CONTINUOUS



- All changes in the field values are gradual
- may quantify, measure the changes
- Ex., may measure slope by the variation of elevation

TYPES OF GEOGRAPHIC PHENOMENA

Fields: CONTINUOUS



TYPES OF GEOGRAPHIC PHENOMENA

Fields: DISCRETE



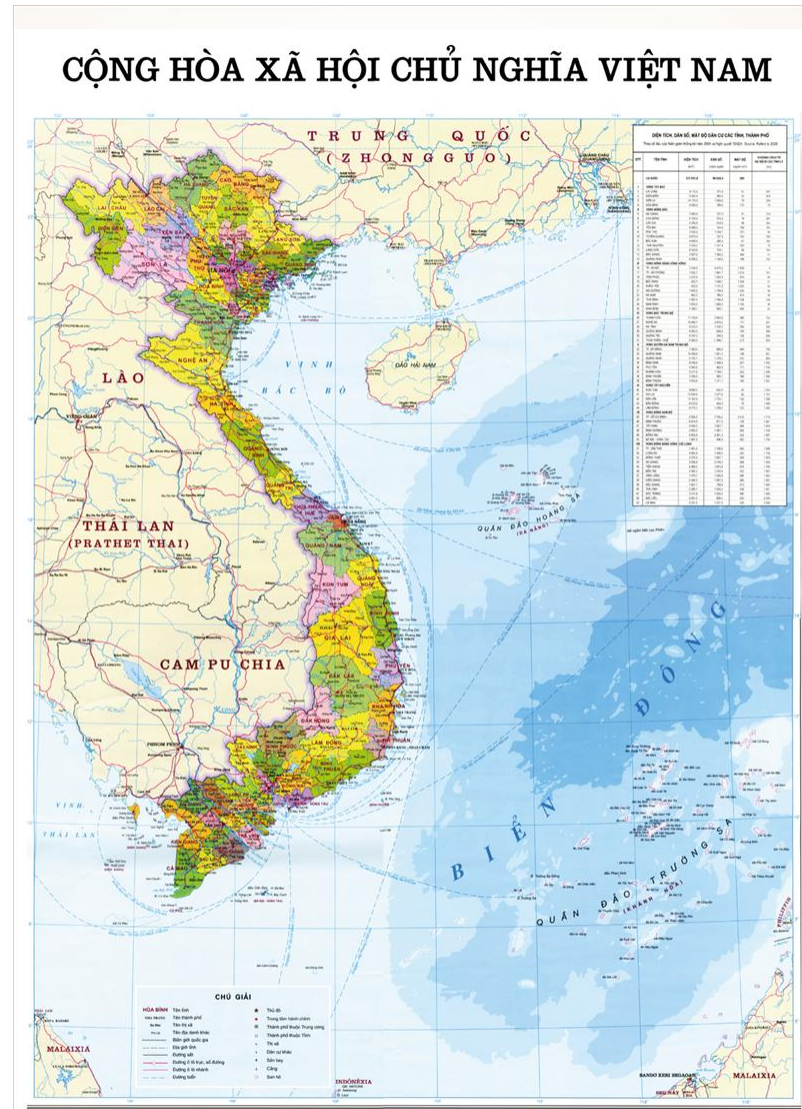
Discrete Fields: separate the study area in subparts

- clear boundary
- values at any points within the boundary are the same
- Ex., land classifications, geological classes, soil types, landuse types, crop types, or vegetation types

Fields: DISCRETE



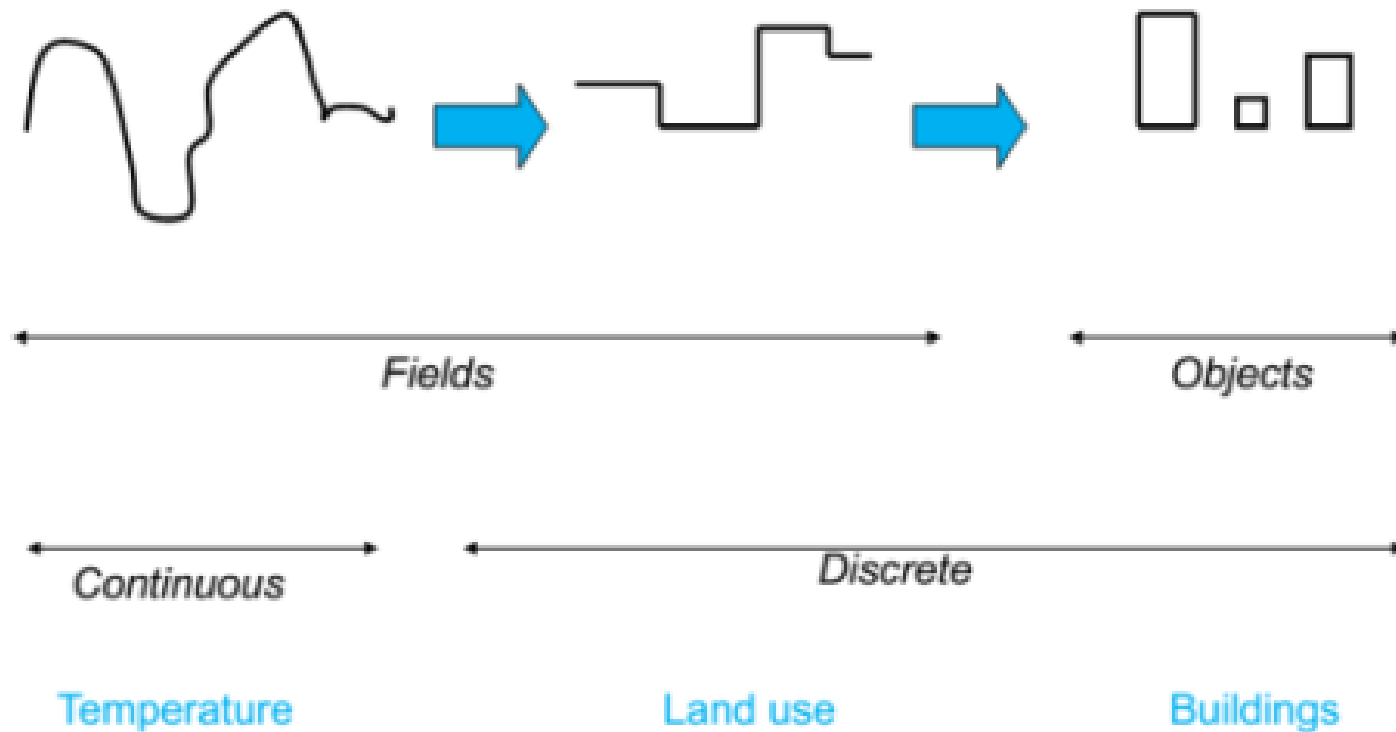
Map of regions of Vietnam



Administration map of Vietnam

TYPES OF GEOGRAPHIC PHENOMENA

Fields: DISCRETE



Geographical features/phenomena

How do we describe geographical features?

- by recognizing two *types of data*:
 - **Spatial data** which describes location (where)
 - **Attribute data** which specifies characteristics at that location (what, how much, and when)

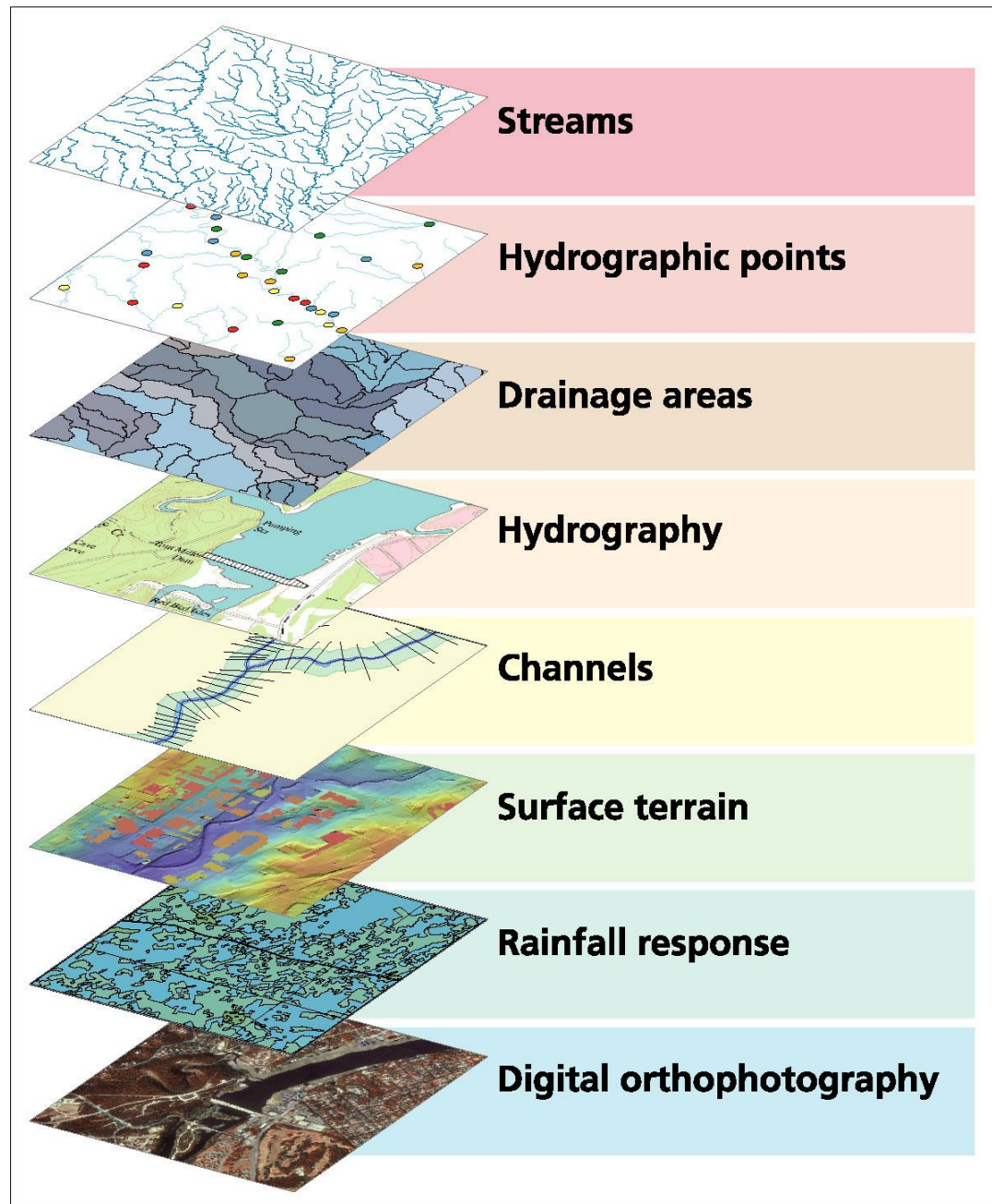
How do we represent these digitally in a GIS?

- by grouping into **layers** based on similar characteristics (e.g hydrography, elevation, water lines, sewer lines, grocery sales) and using either:
 - **vector** data model (*coverage* in ARC/INFO, *shapefile* in ArcView)
 - **raster** data model (*GRID* or *Image* in ARC/INFO & ArcView)
- by selecting appropriate **data properties** for each layer with respect to:
 - **projection, scale, accuracy, and resolution**

How do we incorporate into a computer application system?

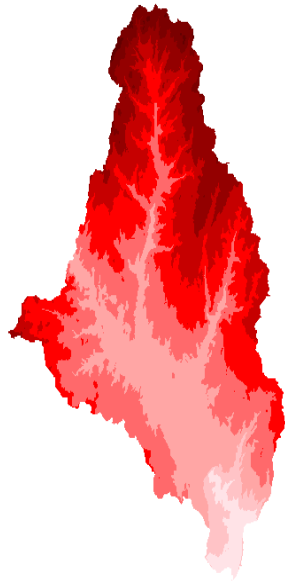
- by using a relational Data Base Management System (DBMS)

GIS Data Model based on data layers or themes

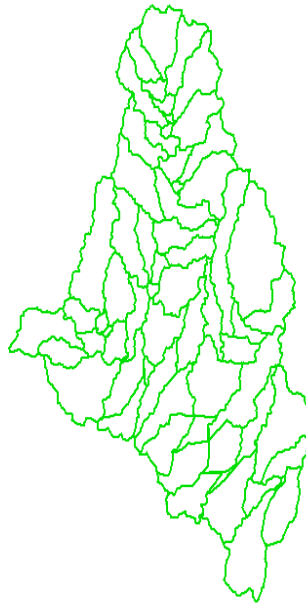


Examples of layers or themes

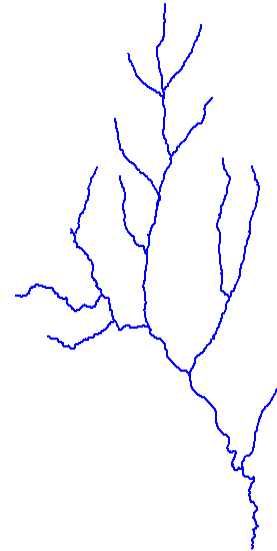
- Data is organized by *layers*, *coverages* or *themes*, with each theme representing a common feature.
- Layers are integrated using explicit location on the earth's surface, *thus geographic location is the organizing principal*.



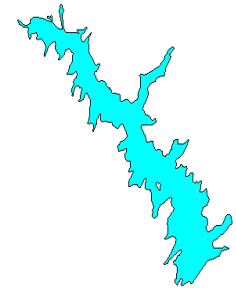
Digital Elevation
Models



Watersheds



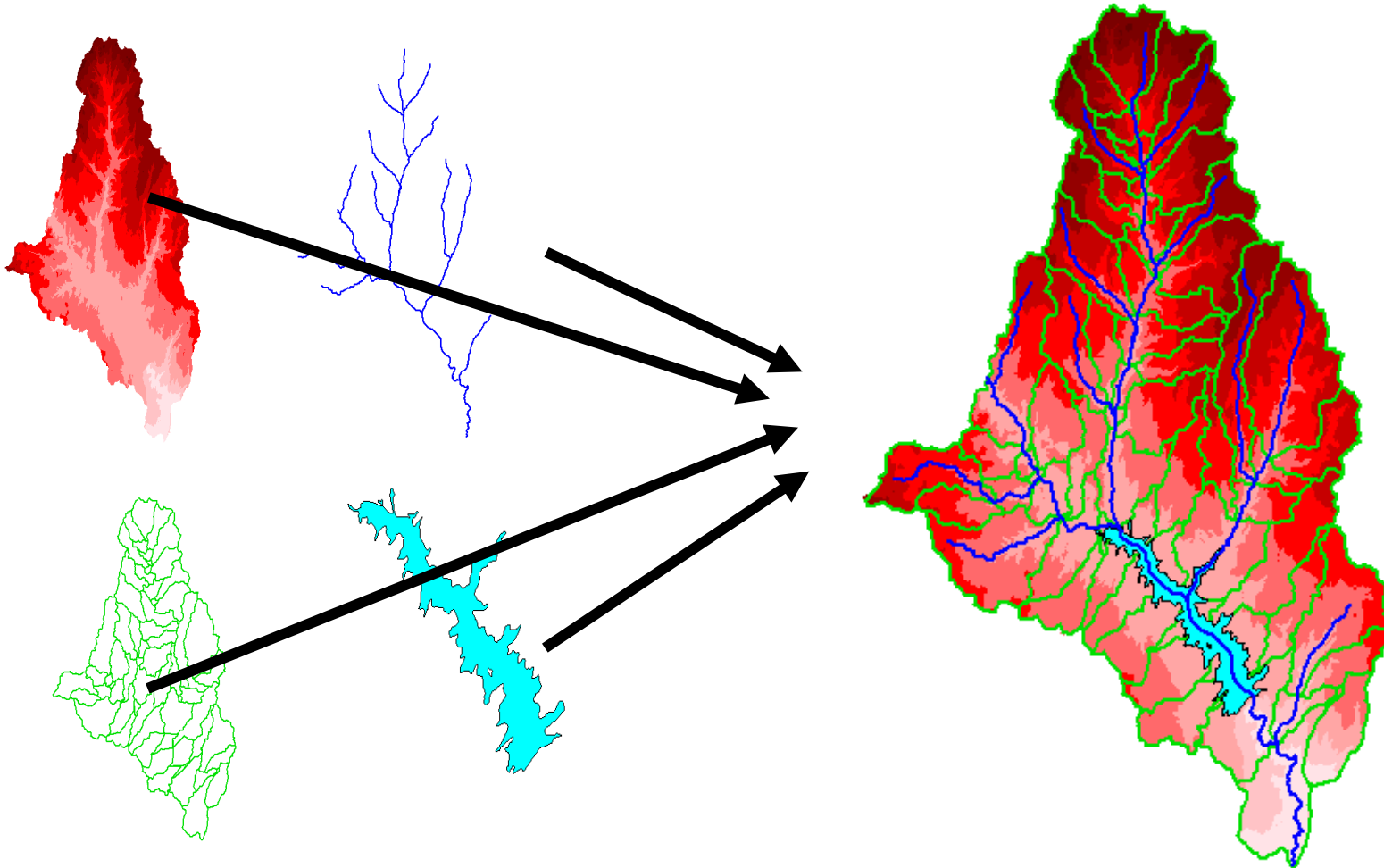
Streams



Waterbodies

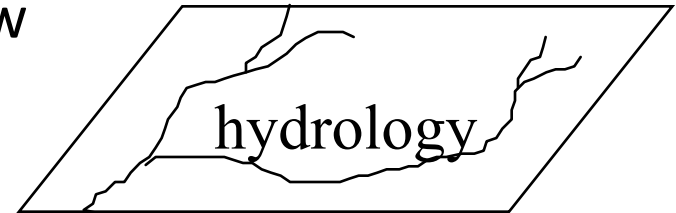
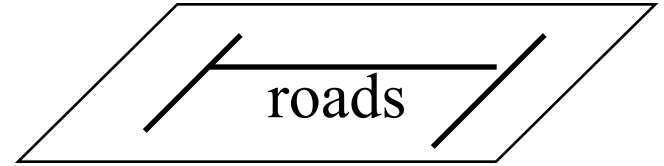
An integrated view

- Layers are integrated using explicit location on the earth's surface, *thus geographic location is the organizing principal.*

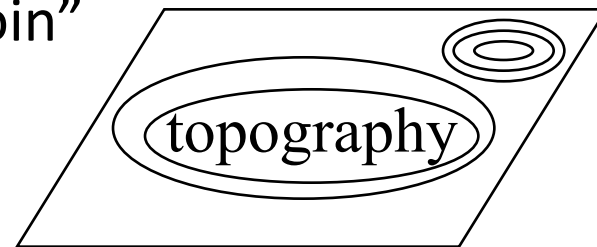


How are layers described?

- Layers are comprised of two data types:
 - *spatial data* which describes location (where)
 - stored in a shape file
 - *attribute data* specifying what, how much, when
 - stored in a database table



GIS systems traditionally maintain spatial and attribute data separately, then “join” them for display or analysis



Attribute data types

Categorical (name):

- nominal
 - no inherent ordering
 - land use types, county names
- ordinal
 - inherent order
 - road class; stream class
- often coded to numbers eg SN but can't do arithmetic

Numerical

Known difference between values

Expressed as *integer* [whole number] or *floating point* [decimal fraction]

- temperature (Celsius or Fahrenheit), income, age, rainfall

Attribute data tables can contain locational information, such as addresses or a list of X,Y coordinates. However, these must be converted to true spatial data (shape file), for example by geocoding, before they can be displayed as a map.

Attribute data types

entity →

Parcel Table			
Parcel #	Address	Block	\$ Value
8	501 N Hi	1	105,450
9	590 N Hi	2	89,780
36	1001 W. Main	4	101,500
75	1175 W. 1st	12	98,000

Key field Attribute

Contain **Tables** or ***feature classes*** in which:

- ***rows***: entities, records, observations, features:
 - ‘all’ information about one occurrence of a feature
- ***columns***: attributes, fields, data elements, variables, items
 - one type of information for all features

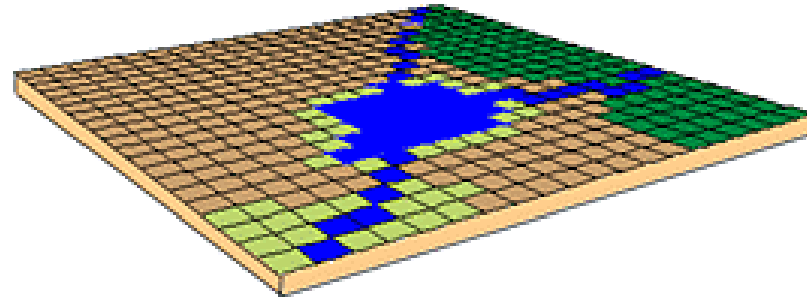
The ***key field*** is an attribute whose values uniquely identify each row

Spatial Data

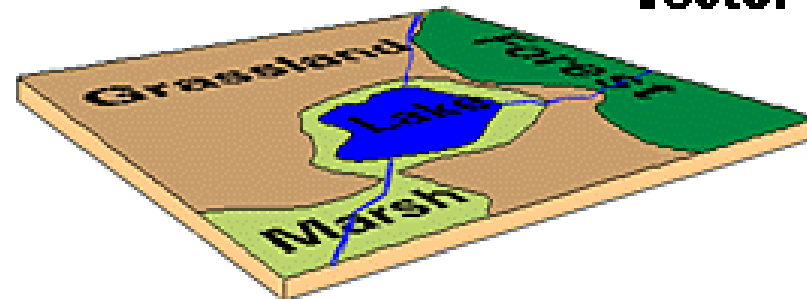
The spatial component of a layer may be represented in two ways:

- in *raster* (*image*) format as pixels
- in *vector* format as points and lines and areas (PLA-model)

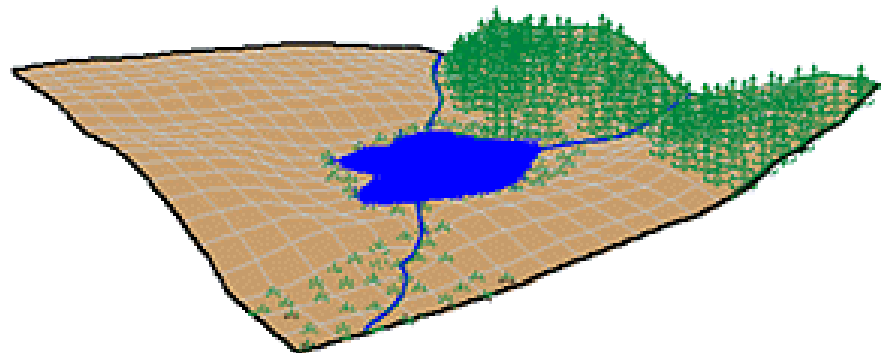
Raster / Image



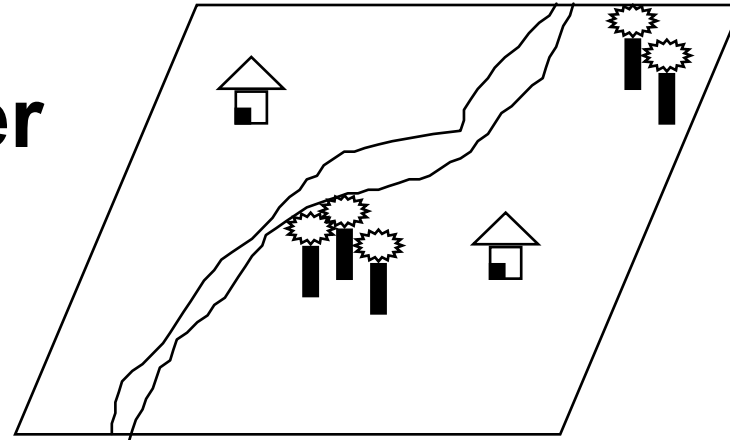
Vector



Real World



Concept of Vector and Raster

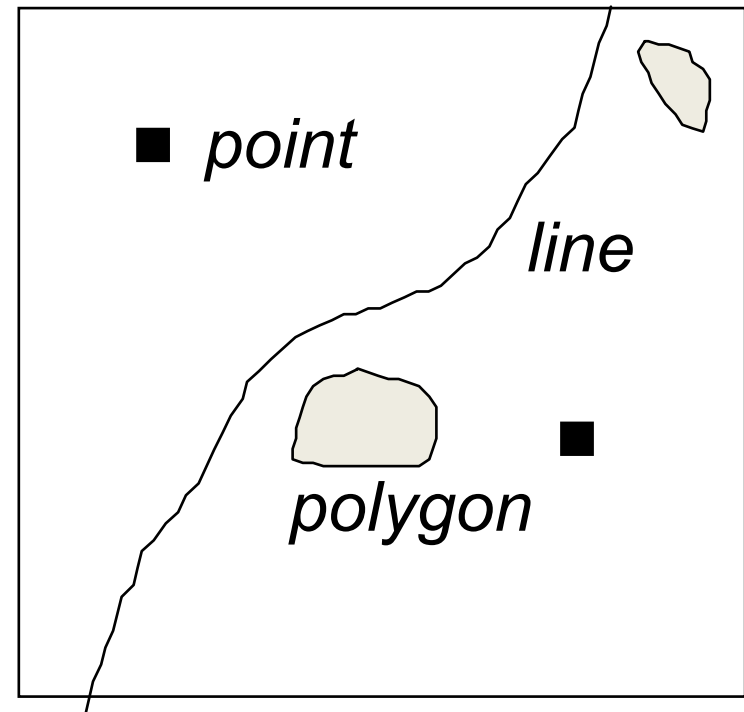


Real World

Raster Representation

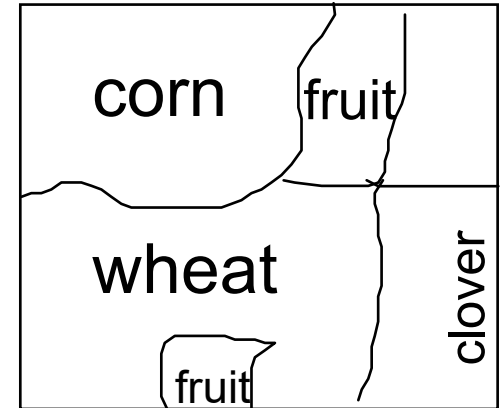
	0	1	2	3	4	5	6	7	8	9
0								R	T	
1							R			T
2		H					R			
3							R			
4					R	R				
5				R						
6			R		T	T		H		
7			R		T	T				
8		R								
9		R								

Vector Representation



Representing Data using *Raster* Model

- Area is covered by grid with (usually) equal-sized cells
- Location of each cell calculated from origin of grid: Column, row
- Cells often called *pixels* (picture elements); raster data often called *image* data
- Attributes are recorded by assigning each cell a single value based on the majority feature (attribute) in the cell, such as land use type.
- Easy to do overlays/analyses, just by 'combining' corresponding cell values: "*yield = rainfall + fertilizer*" (why raster is faster, at least for some things)
- Simple data structure: directly store each layer as a single table



	0	1	2	3	4	5	6	7	8	9
0	1	1	1	1	1	4	4	5	5	5
1	1	1	1	1	1	4	4	5	5	5
2	1	1	1	1	1	4	4	5	5	5
3	1	1	1	1	1	4	4	5	5	5
4	1	1	1	1	1	4	4	5	5	5
5	2	2	2	2	2	2	2	3	3	3
6	2	2	2	2	2	2	2	3	3	3
7	2	2	2	2	2	2	2	3	3	3
8	2	2	4	4	2	2	2	3	3	3
9	2	2	4	4	2	2	2	3	3	3

Raster Data Structures

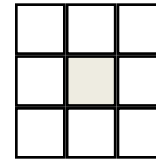
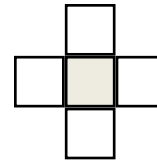
- **Square grid:** equal length sides

- 4-connected neighborhood (*rook's case*)

- all neighboring cells are equidistant

- 8-connected neighborhood (*queen's case*)

- all neighboring cells **not** equidistant



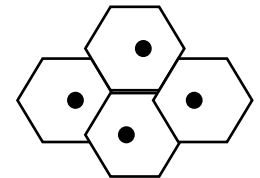
- **Rectangular**

- commonly occurs for lat/long when projected

- data collected at 1degree by 1 degree will be varying sized rectangles

- **triangular** (3-sided) and **hexagonal** (6-sided)

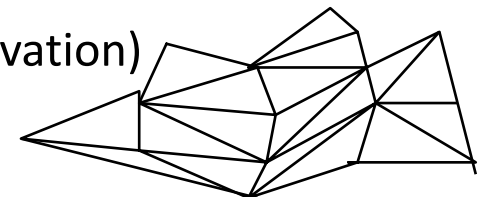
- all adjacent cells and points are equidistant



- **triangulated irregular network (TIN):**

- vector* model used to represent continuous surfaces (elevation)

- more later under vector



Raster Data Structures

Runlength Compression (for single layer)

Full Matrix--162 bytes

111111122222222223
111111122222222233
111111122222222333
111111222222223333
111113333333333333
111113333333333333
111113333333333333
111333333333333333
111333333333333333

Run Length (row)--44 bytes

1,7,2,17,3,18

1,7,2,16,3,18

1,7,2,15,3,18

1,6,2,14,3,18

1,5,3,18

This is a “*lossless*”
compression, as
opposed to “*lossy*,”
since the original
data can be exactly
reproduced. (*Indicate*

1,5,3,18

1,5,3,18

1,3,3,18

1,3,3,18

*the values and the end
location of the value in
each row)*

Raster Model

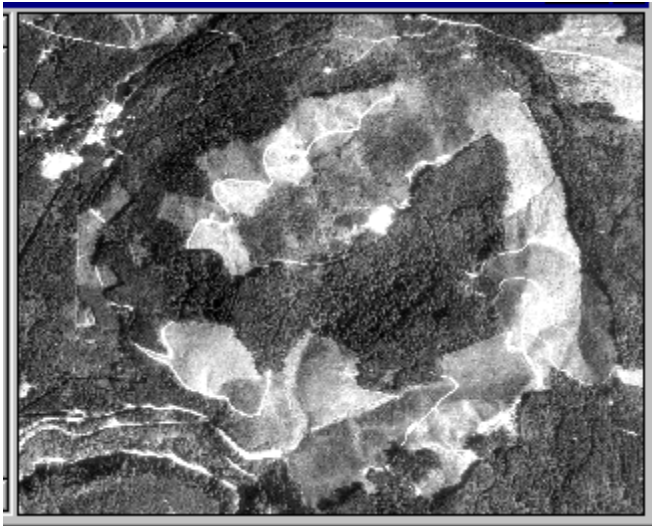
Raster data are good at representing continuous phenomena, e.g.,

- Wind speed
- Elevation, slope, aspect
- Chemical concentration
- Likelihood of existence of a certain species
- Electromagnetic reflectance (photographic or satellite imagery)

Raster Model

Much data comes in this form

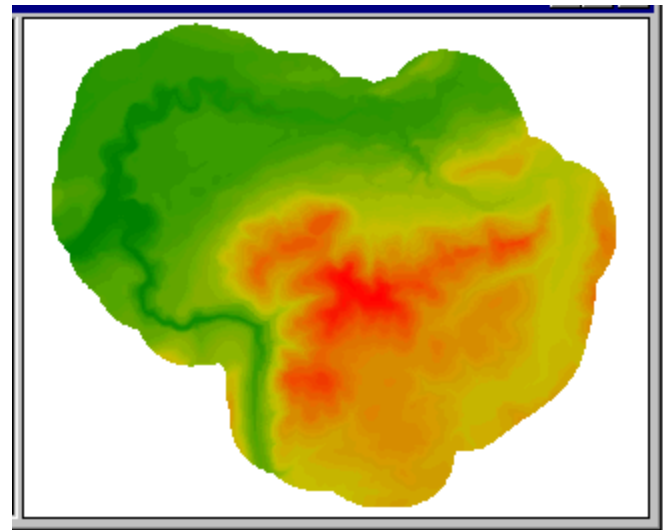
- images from remote sensing (LANDSAT, SPOT)
- scanned maps



- digital orthophotos

Best for continuous features:

- elevation
- temperature
- soil type
- land use



- digital elevation model (DEM)

Raster Model: Pros and Cons

- [+] Continuous (surface) data represented easily
- [+] Simple data structure, fast indexing
- [–] Shape of discrete polygonal features generalized by cells
- [–] Intersection of two lines

Vector Format

- point (node): 0-dimension
 - single x,y coordinate pair
 - zero area
 - tree, oil well, label location
- line (arc): 1-dimension
 - two (or more) connected x,y coordinates
 - road, stream
- polygon : 2-dimensions
 - four or more ordered and connected x,y coordinates
 - first and last x,y pairs are the same
 - encloses an area
 - census tracts, county, lake

Point

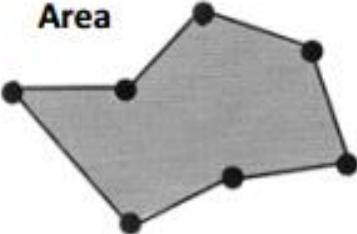
 (x, y)

Lines

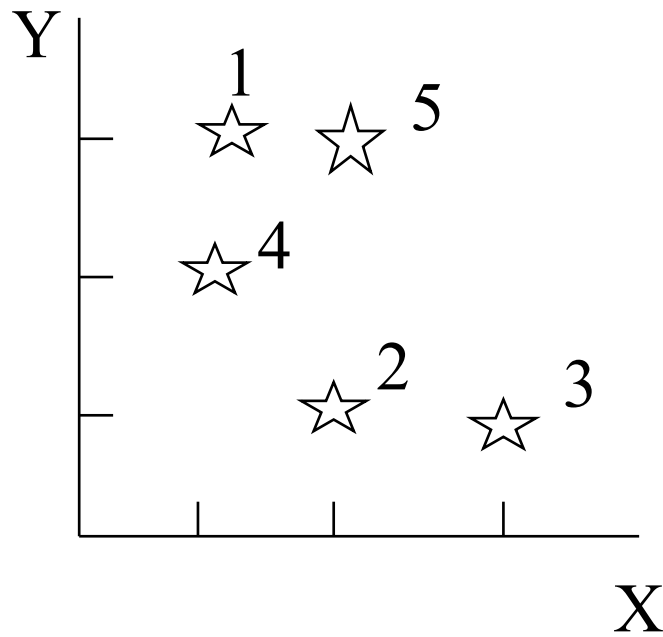
 $(x_1, y_1), (x_2, y_2)$

Curve

 $(x_1, y_1), \dots, (x_n, y_n)$

Area

 $(x_1, y_1), \dots, (x_{n-1}, y_{n-1}), (x_1, y_1)$

Point Data using the *Vector Model*: *data implementation*



- Features in the theme (coverage) have unique identifiers--point ID, polygon ID, arc ID, etc

- common identifiers provide link to:

- coordinates table (for ‘where’)

- attributes table (for what)

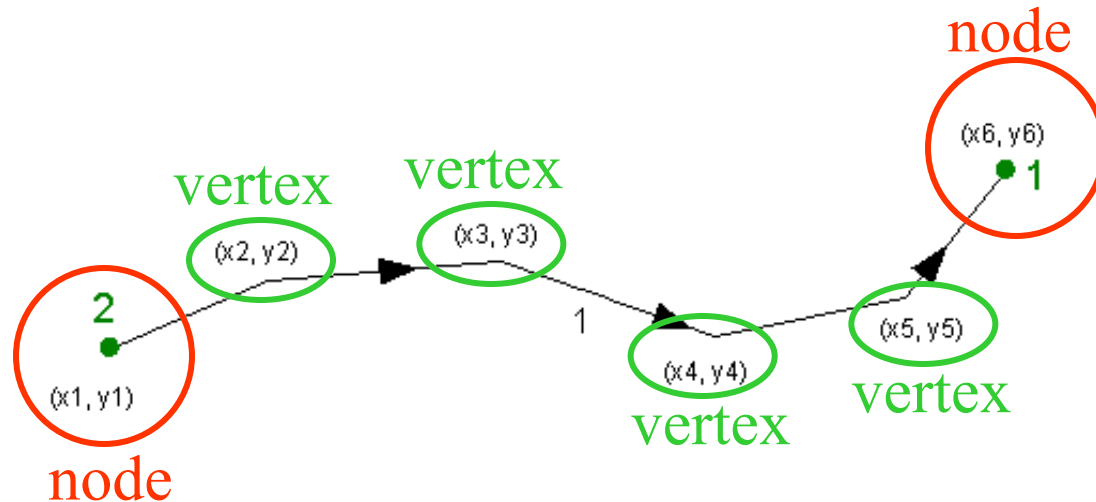
Coordinates Table		
Point ID	x	y
1	1	3
2	2	1
3	4	1
4	1	2
5	3	2

Attributes Table		
Point ID	model	year
1	a	90
2	b	90
3	b	80
4	a	70
5	c	70

- Again, concepts are those of a relational data base, which is really a prerequisite for the vector model

Vector Model

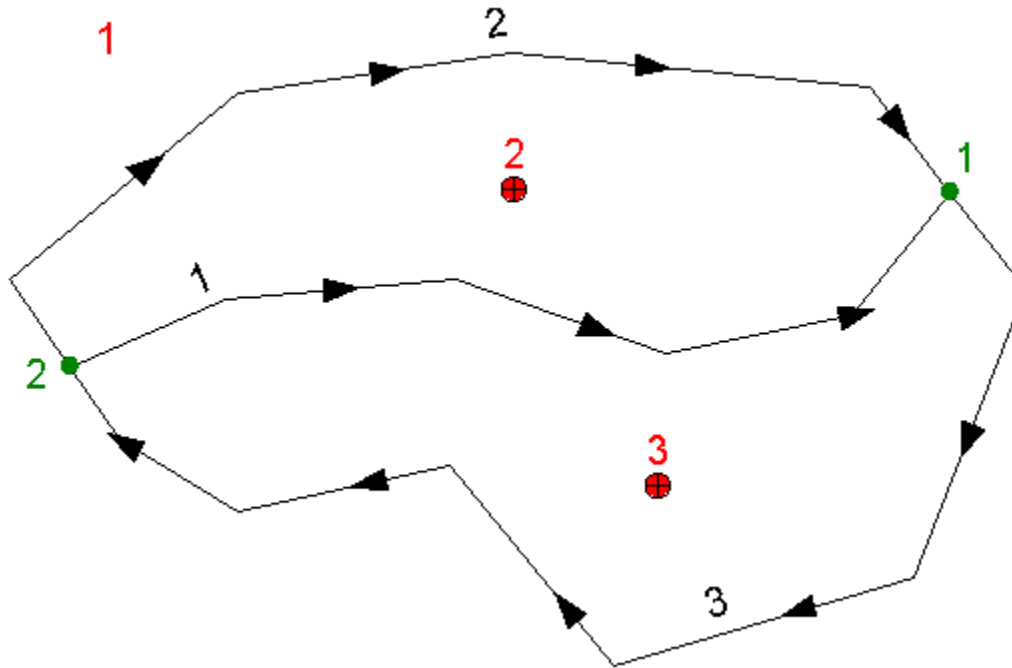
Lines: structure



- Lines start and end at nodes
 - line #1 goes from node #2 to node #1
- Vertices determine shape of line
- Nodes and vertices are stored as coordinate pairs

Vector Model

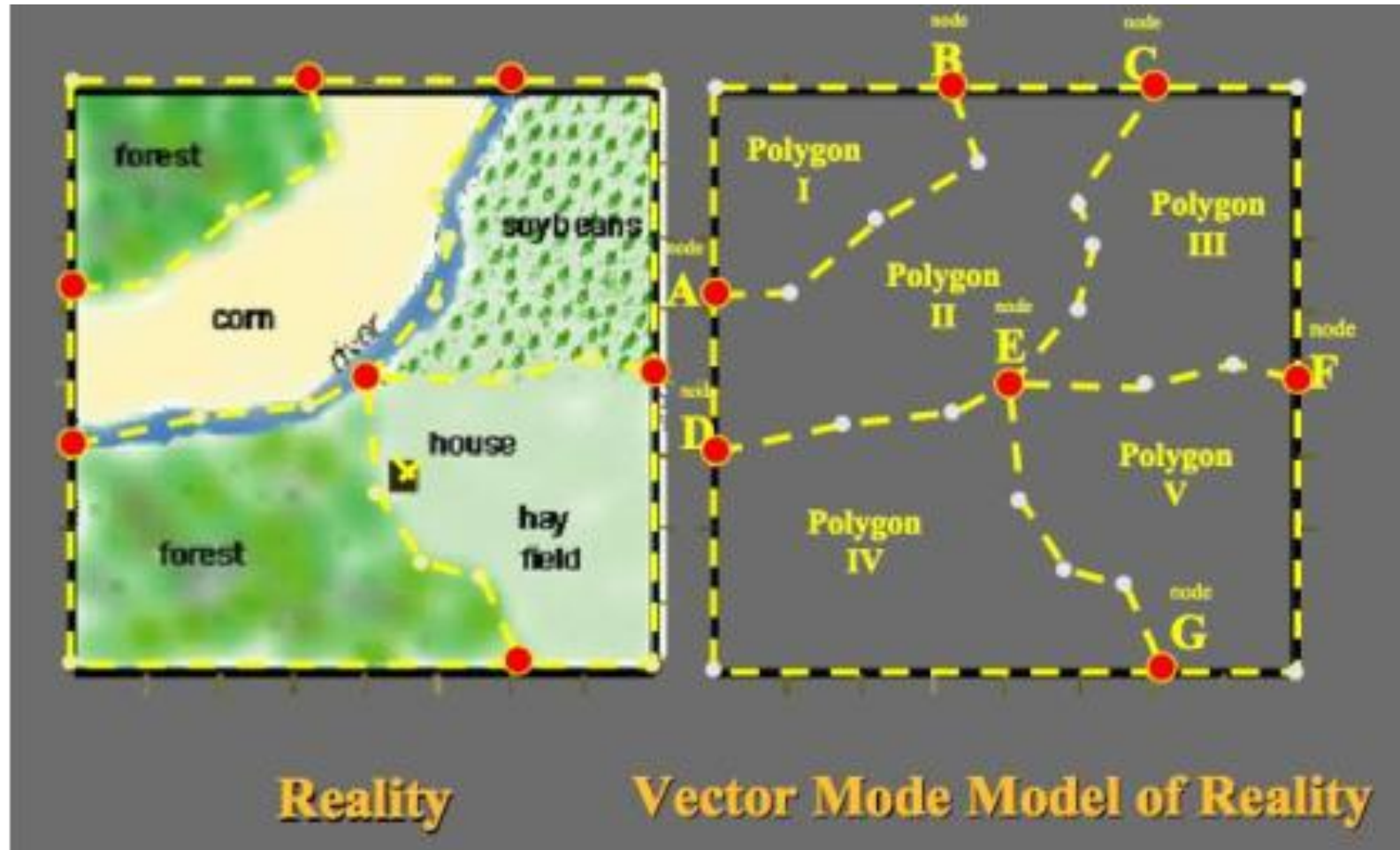
Polygons: structure

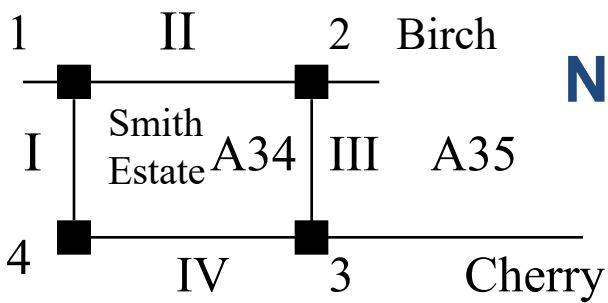


- complex data model, especially for larger data sets
- arc-node topology

Vector Model

Polygons: fundamental spatial data model





Node/Arc/ Polygon and Attribute Data

Relational Representation: DBMS required!

Spatial Data

Node Table

Node ID	Easting	Northing
1	126.5	578.1
2	218.6	581.9
3	224.2	470.4
4	129.1	471.9

Arc Table

Arc ID	From N	To N	L Poly	R Poly
I	4	1		A34
II	1	2		A34
III	2	3	A35	A34
IV	3	4		A34

Polygon Table

Polygon ID	Arc List
A34	I, II, III, IV
A35	III, VI, VII, XI

Attribute Data

Node Feature Attribute Table

Node ID	Control	Crosswalk	ADA?
1	light	yes	yes
2	stop	no	no
3	yield	no	no
4	none	yes	no

Arc Feature Attribute Table

Arc ID	Length	Condition	Lanes	Name
I	106	good	4	
II	92	poor	4	Birch
III	111	fair	2	
IV	95	fair	2	Cherry

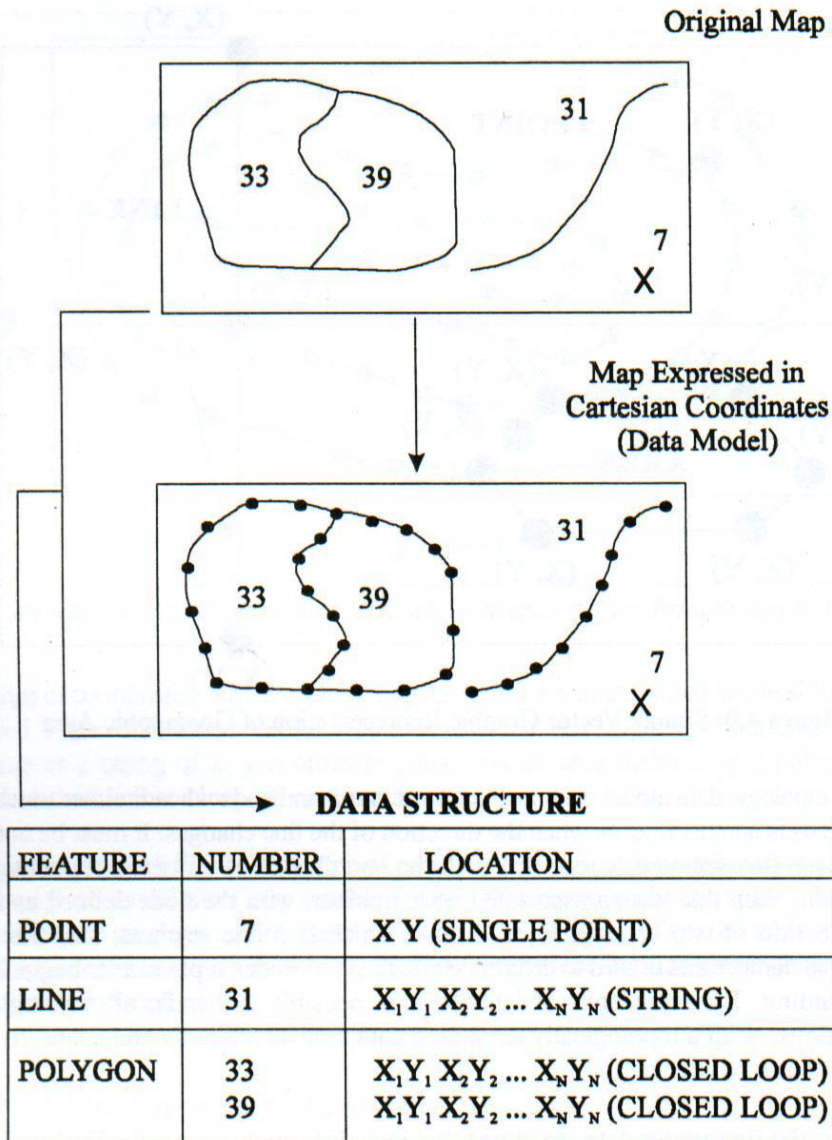
Polygon Feature Attribute Table

Polygon ID	Owner	Address
A34	J. Smith	500 Birch
A35	R. White	200 Main

Variety of Vector Models

- Spaghetti model
- Topological model (most common)
- Triangulated irregular network (TIN)

Vector Model: Spaghetti

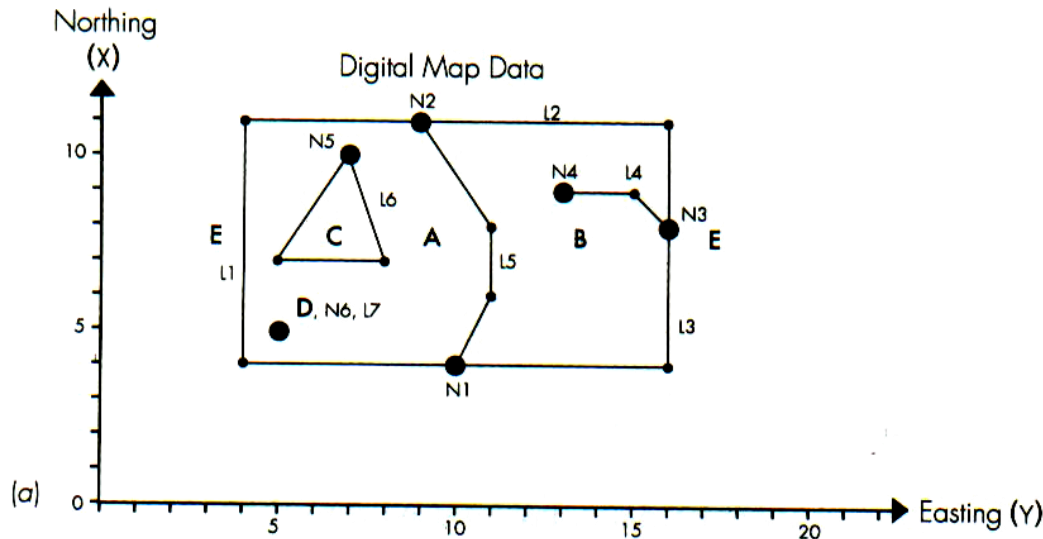


- Very efficient algorithms to detect properties
- Difficult to calculate in case of overlaying polygons

Source: Lakhan, V. Chris. (1996).
Introductory Geographical Information Systems. p. 54.

Figure 4.10: The “Spaghetti” Data Model. Source: Adapted from drawing presented by Dangermond (1983).

Vector Model: Topological



The **topological data model** is used for four relations

- R1: every line has two endpoints
- R2: every line has two areas
- R3: every area is surrounded by lines
- R4: every point is surrounded by areas and lines

(b) POLYGON TOPOLOGY

Polygon	Links
A	L1, L5
B	L2, L3, L5
C	L6
D	L7
E	L1, L2, L3

(c) NODE TOPOLOGY

Node	Links
N1	L1, L3, L5
N2	L1, L2, L5
N3	L2, L3, L4
N4	L4
N5	L6
N6	L7

(d) LINK TOPOLOGY

Links	Start node	End node	Left polygon	Right polygon
L1	N1	N2	E	A
L2	N2	N3	E	B
L3	N3	N1	E	B
L4	N3	N4	B	B
L5	N2	N1	B	A
L6	N5	N5	A	C
L7	N6	N6	A	A

(e) LINK COORDINATES

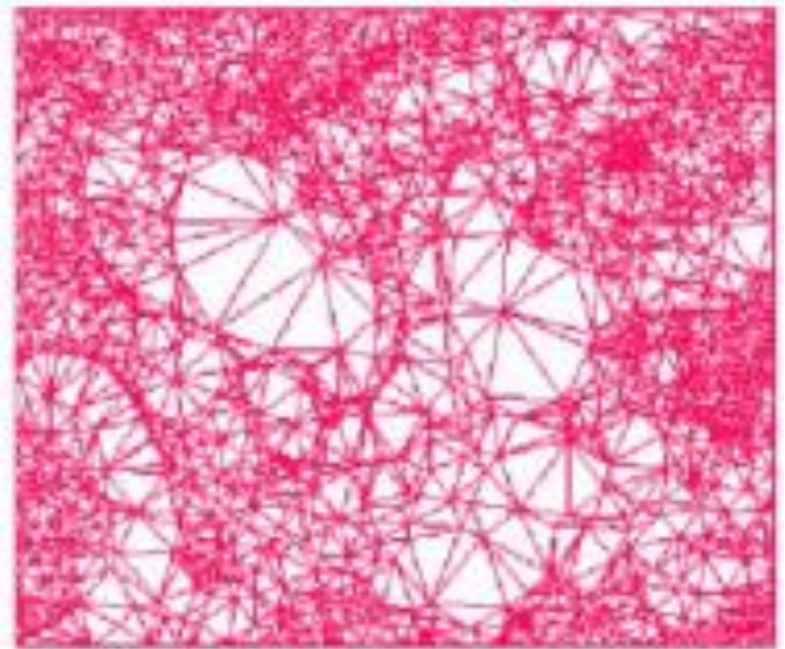
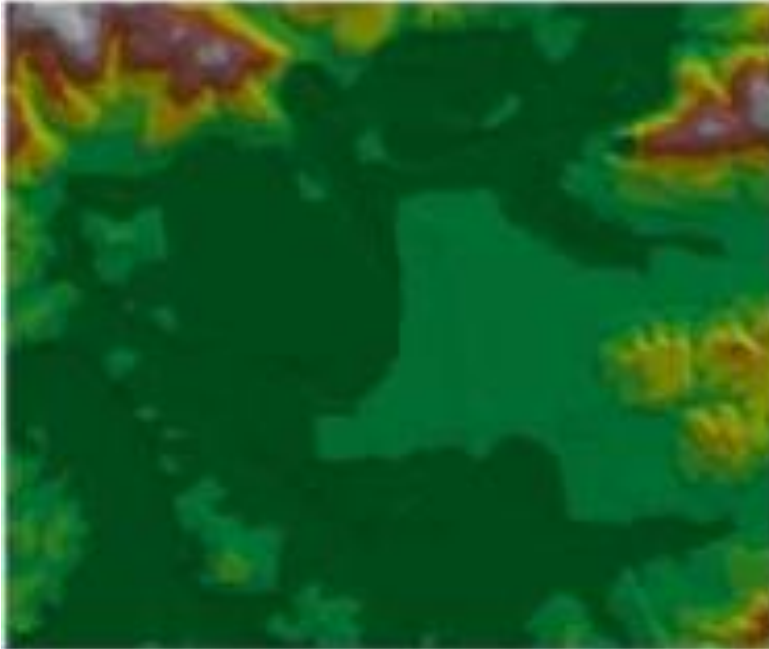
Link	Coordinates			
L1	4,10	4,4	11,4	11,9
L2	11,9	11,16	8,16	
L3	8,16	4,16	4,10	
L4	8,16	9,15	9,13	
L5	11,9	8,11	6,11	4,10
L6	10,7	7,8	7,5	10,7
L7	5,5			

Topological Data Model

The **topological data model** is used four relations

- R1: every line has two endpoints
- R2: every line has two areas
- R3: every area is surrounded by lines
- R4: every point is surrounded by areas and lines

TIN



Advantage and Disadvantage of using raster and vector Data

<https://grindgis.com/what-is-gis/what-is-gis-definition>

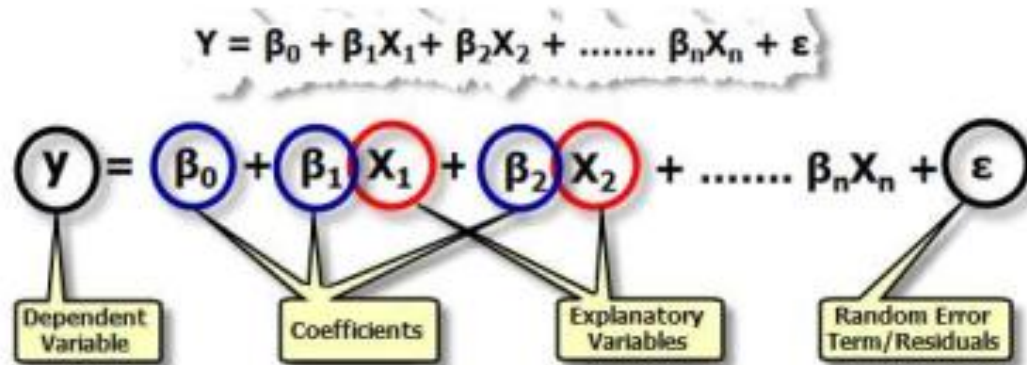
- Raster data model record value of all the points of the area covered which required more data storage than model represented by the vector model.
- Raster data is less expensive to create computationally compare to vector graphics.
- Raster data has issue while overlaying multiple images.
- Vector data are easily overlaid, for example overlaying roads, rivers, land use are easier than raster data.
- Vector data are easier to scale, re-project or register.
- Vector data are more compatible with the relational database management system.
- Vector file sizes are way smaller than raster image file sizes.
- Vector data are easier to update like adding river stream but has to be recreated for the raster image.

4. Regression models and Process Models

- Regression model relates a dependent variable to a number of independent (explanatory) variables in an equation which can then be used for prediction or estimation
- Regression model can use an overlay operation in GIS to combine variables needed for the analysis

Linear Regression Model

<u>y</u>	=	<u>$\alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_i x_i$</u>
Dependent		Independent variables
Predicted		Predictor variables
Response variable		Explanatory variables
Outcome variable		Covariables



The primary purpose of linear regression is to predict values of **Y** from values of **X_i**

Regression model applications

- Modeling traffic accidents as a function of speed, road conditions, weather, and so forth, to inform policy aimed at decreasing accidents.
- Modeling property loss from fire as a function of variables such as degree of fire department involvement, response time, or property values.
- Measuring the extent that changes in one or more variables jointly affect changes in another. Example: Understand the key characteristics of the habitat for some particular endangered species of bird (perhaps precipitation, food sources, vegetation, predators) to assist in designing legislation aimed at protecting that species.
- It is mainly applied for bird habitat identification, rainfall triggered land slide model, predicting grass land bird habitat attitude towards national park designation

Process Model

- A process model: represent an evolution, integrates existing knowledge about the environment process in the real world into: a set of relationships and equations for quantifying the processes
- A process model offers both a predictive capability and an explanation that is inherent in the proposed processes
- Process models are by definition **predictive** and **dynamic** models
- Environmental models are very complex and data intensive
- Environmental models are typically process models because they must deal with the interaction of many variables including physical variables such as climate, topography, vegetation, and soils as well as cultural variables such as land management, run dynamic weather patterns, storms, and hurricanes

Revised Universal Soil Loss Equation (RUSLE)

- RUSLE is a model that is widely used to estimate average annual nonchannelized soil loss.
- Soil erosion is an environmental Process that involves climate, soil properties, topography, soil surface conditions and human activities
- A well known model of soil erosion is the Revisited Universal Soil Loss Equation (RUSLE)
- RUSLE predicts the average soil loss carried by runoff from specific field slopes in specified cropping and management systems from range land

RUSLE is a multiplicative model

- with six factors

$$A = R * K * L * S * C * P$$

Where A is average soil loss

R- is the rainfall run off erosivity factor

K is the soil erodibility factor

L is the slope length factor

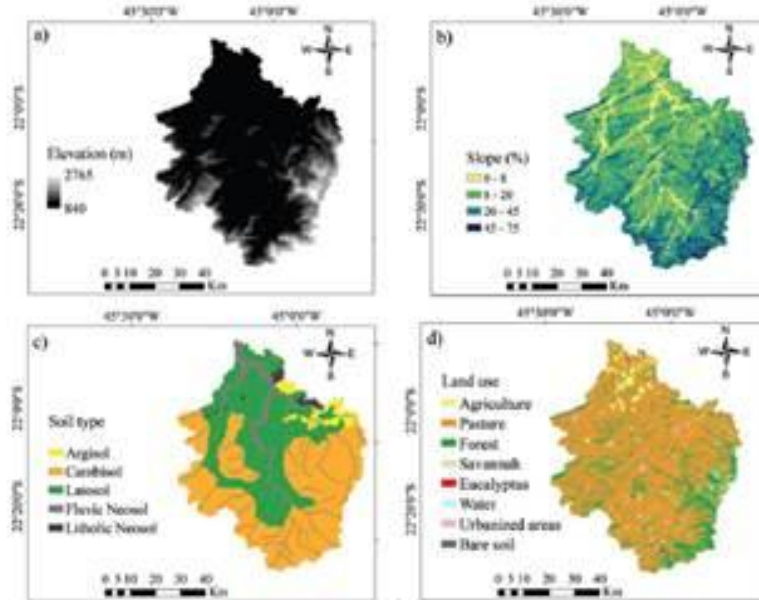
S is the slope steepness factor

C is crop management factor(land cover) and

P = Support practice factor (conservation)

Example of RUSLE model

(1)



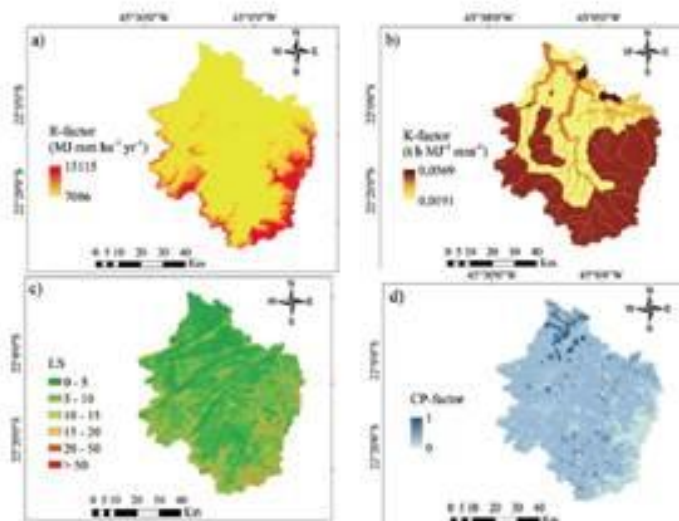
(1) Basic maps for running RUSLE

(2) RUSLE factors maps derived from the basic maps

(3) Algebra map to combine of factors maps

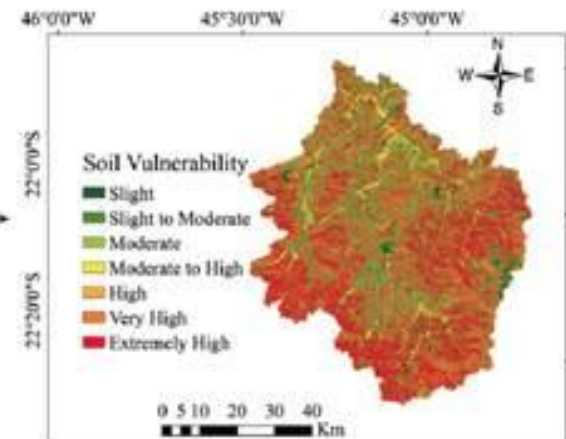
(4) Soil erosion risk map

(2)



(3)

(4)



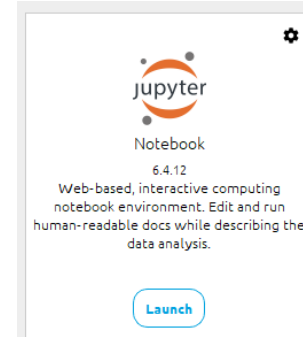
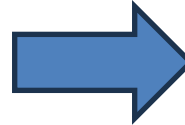
Practice 1. Raster and vector

(Create a vector file with Python)

1. Practices using Python

Download and install Anaconda/Jupyter notebook

(<https://www.anaconda.com/download>)



2. Create a “New Python 3”.

#Install library:

```
pip install geopandas
```

#Call library:

```
import pandas as pd
```

```
import geopandas as gpd
```

```
from shapely.geometry import Point, LineString, Polygon
```

```
import fiona
```

```
import matplotlib.pyplot as plt
```

#2.1 input and print a point

```
coordinate = [105.801228, 21.048171]
```

```
point_coord = Point(coordinate)
```

```
pair
```

```
df = {'USTH': ['Dept Space and Applications'], 'geometry': [point_coord]}  
required geometry column
```

```
point = gpd.GeoDataFrame(df, geometry='geometry', crs = "EPSG:4326")
```

#coordinate of

Create a Shapely point from a coordinate

generate a dataframe with attributes and

Convert shapely object to a geodataframe

```
plt.plot ()
```

```
plt.scatter (105.801228, 21.048171)
```

#points

```
plt.annotate (u'Dept Space and Applications',(104, 20.8), color='k', fontsize=11,rotation=0 ) #text
```

```
point.plot(column='USTH', legend=True) # Print field 'USTH' with legend
```

Practice 1. Raster and vector

(Create a vector file with Python)

#2.2 List of attributes and coordinates

```
df = pd.DataFrame(  
    {'building': ['USTH', 'STI', 'VNSC', 'IET', 'Center building'],  
     'Name': ['A21', 'A22', 'A6', 'A13', 'A3'],  
     'lat': [105.801228, 105.802043, 105.802045, 105.800214, 105.799930],  
     'lon': [21.048171, 21.048092, 21.046966, 21.048490, 21.046829]})  
  
ply_coord = [Point(x, y) for x, y in zip(df.lat, df.lon)]           # Generate points from the coordinate list  
  
poly = gpd.GeoDataFrame(df, geometry=ply_coord, crs = "EPSG:4326")  # Convert objects to a geodataframe with a crs  
  
poly.plot(column='building', legend=True)                          # Plot map
```

#2.3 Import a table to map

```
from io import StringIO  
data = """  
ID,X,Y,floor  
1, 105.801228, 21.048171, 10  
1, 105.802043, 21.048092, 8  
2, 105.802045, 21.046966, 12  
2, 105.800214, 21.048490, 4  
2, 105.799930, 21.046829, 16  
"""  
  
df = pd.read_table(StringIO(data), sep=',')           # use StringIO to read in text chunk  
  
points = [Point(xy) for xy in zip(df.X, df.Y)]        #zip the coordinates into a point object and convert to a GeoData Frame  
  
geo_df = gpd.GeoDataFrame(df, geometry=points, crs = 'EPSG:4326')  
geo_df.plot()
```

Practice 1. Raster and vector

(Create a vector file with Python)

#2.4 Generate lines from points

```
lines = geo_df.groupby(['ID'])['geometry'].apply(lambda x: LineString(x.tolist()))    # each `ID` group of points as a line
```

```
lines = gpd.GeoDataFrame(lines, geometry='geometry', crs="EPSG:4326")    # store as a GeodataFrame and add 'ID' as a column (currently stored as the 'index')
lines.reset_index(inplace=True)
lines.plot(column='ID')
```

#2.5 Generate a polygon from points and add it to map

```
coordinates = [[105.801228, 21.048171], [105.802043, 21.048092], [105.802045, 21.046966], [105.799930, 21.046829], [105.800214, 21.048490]]
```

```
# Create a polygon using the list of coordinates
ply_coord = Polygon(coordinates)
```

```
# Create a dictionary to hold the necessary attributes and geometry column
df = {'Attribute': ['name1'], 'area': ['1 ha '], 'geometry': [ply_coord]}
```

```
# Convert the object into a Geodataframe and assign a CRS
poly = gpd.GeoDataFrame(df, geometry='geometry', crs="EPSG:4326")
```

```
poly.plot()
```

#2.6 Export the polygon to shape file

```
outfp = r"~\AppData\Local\Temp\Vast.shp"    # Specify the output path for the Shapefile
poly.to_file(outfp)    # Export the data to the specified Shapefile
```

Practice 1. Raster and vector

(Working with GIS software)

1. Introduction about QGIS

Download (<https://qgis.org/en/site/forusers/download.html>)

Please download and install the “Long term release” version below



2. Open raster file
3. Read information of the raster
4. Subtract the raster by an extent
5. Design a geographic theme and attribute information
6. Create a point layer with attribute information
7. Create a line layer with attribute information
8. Create a polygon layer with attribute information

Practice to create a land cover map

1. Digitalize land covers

Digitalizing fully the image with a land cover layer (Land cover layer with fields: Types, Owner, Area, Price, Total_price, “Residence, Fish ponds, Annual agriculture, Roads, Cemetery”).

A line layer (Electric lines layer with Fields: Power, Material)

A point layer (Tree layers with Fields: Types, Height, Planted_year, Ages.).

Layer/ Create layer/ New shapefile layer (Select Geometry, Coordinates, Add fields)

2. Add attribute information

Add appropriate information

3. Editing a land cover map

Working with “Layout”

Creating: Legend, North arrow, Scale bar, Grid

Project/ New print layout

4. Converting raster and vectors

Raster/ Conversion/ Rasterizer(Vector to Raster)

DEFINITION OF LAND COVER, LAND COVER CLASSIFICATION

- Land cover is the natural/artificial compositions at the surface

*Land cover data documents **how much** of a region is **covered** by forests, wetlands, impervious surfaces, agriculture, and other land and water types. Water types include wetlands or open water*

*Each group of **above object** is defined as a **class** in land cover map.*

- Land cover classification

*Classification is the “**ordering or arrangement** of objects into **groups or sets** on the basis of their relationships” (Sokal, 1974). It is an abstract representation of the **situation in the field**, using well-defined diagnostic **criteria**.*

- Classification scheme = land cover classification system
- Images (Satellite images, Aerial photos)
- Interpretation key = class samples
- Methods for classification